

Evaluating Fantasy Football Player & Team Statistics

Contents

#	Section	Target Insight	Key Findings
1	Data & Setup	What data, scoring format and features are we working with?	
2	Drivers of Fantasy Performance	Which statistics correlate most strongly with fantasy points?	The Most Important Statistics by Scoring Format
3	Position-Specific Drivers	How do the different positions relate to fantasy football performance?	Positional Importance Differs by Scoring Format
4	Positional Draft Strategy	How should I prioritize selecting different positions?	Ranking Positions by Draft Priority
5	Player Selection by Position	How do the statistics relate to each other, and which ones identify strong players at each position?	Selecting QBs Selecting RBs Selecting WRs Selecting TEs
6	Key Findings Summary & Recommendations	What matters most, what's reliable, and what are the limitations?	

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RendererRegistry.enable('html')
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Introduction

Data & Setup

Load Data

The grain of the data is two fold, NFL data is one row per player per week, while the college data is a season aggregate. As a result, we will need to primarily focus on returning NFL players when evaluating efficiency, which we will represent in points per week.

```
Column Count: 162
Row Count: 166648
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1	00-0016919	A.Vinatieri	Adam Vinatieri	K	SPEC	https://static.www.nfl.com/image/priv
2	00-0019596	T.Brady	Tom Brady	QB	QB	https://static.www.nfl.com/image/priv
3	00-0019646	S.Janikowski	Sebastian Janikowski	K	SPEC	https://static.www.nfl.com/image/priv
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5 rows × 162 columns

Drivers of Fantasy Performance

This section looks at which variables correlate most strongly with fantasy points.

Due to the limitations of the college data, this will be focused on returning NFL Players.

- Players are intentionally not split by position here: the goal is to maximize points per week, not just to find position-specific patterns.
- Pooling all positions shows which kinds of offensive production generate points most efficiently, and so which positions are the most efficient scorers overall.

Definition: Scoring Type

- PPR (points per reception): players earn 1 point for every catch, on top of yardage and touchdown points.
- Non-PPR (standard): catches earn no points on their own; only yardage and touchdowns count.

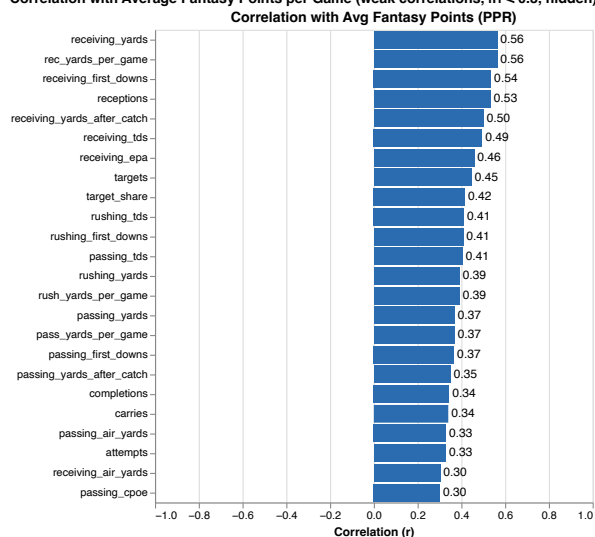
Throughout this analysis we will be examining the data in both contexts as this single rule can drastically affect how the game is played optimally.

Correlations

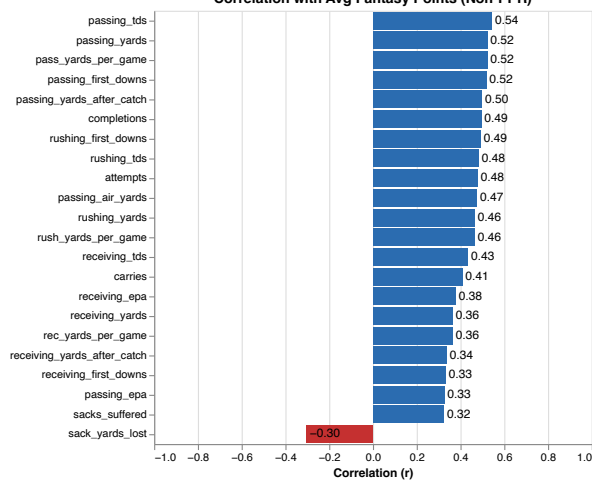
The following visuals highlight moderate and strong correlations between statistics and player performance.

In this first pass **we have not separated by position intentionally**. Before examining position specific performance drivers, we want to uncover high level trends which will provide the foundational strategy for selecting players.

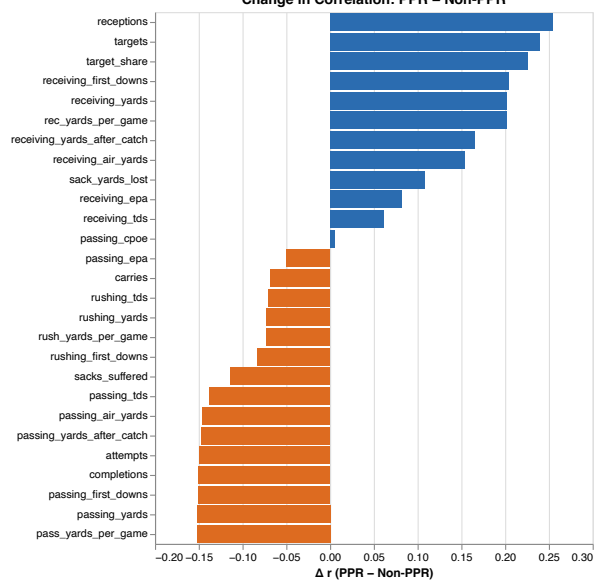
Correlation with Average Fantasy Points per Game (weak correlations, |r| < 0.3, hidden)



Correlation with Avg Fantasy Points (Non-PPR)



Change in Correlation: PPR - Non-PPR

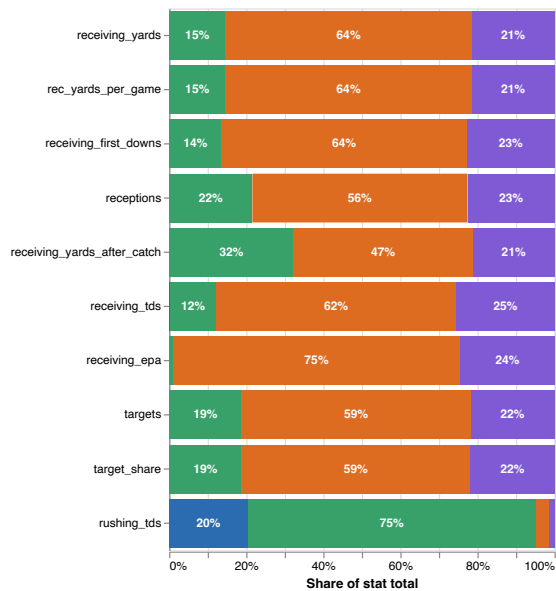


Which positions do these stats belong to?

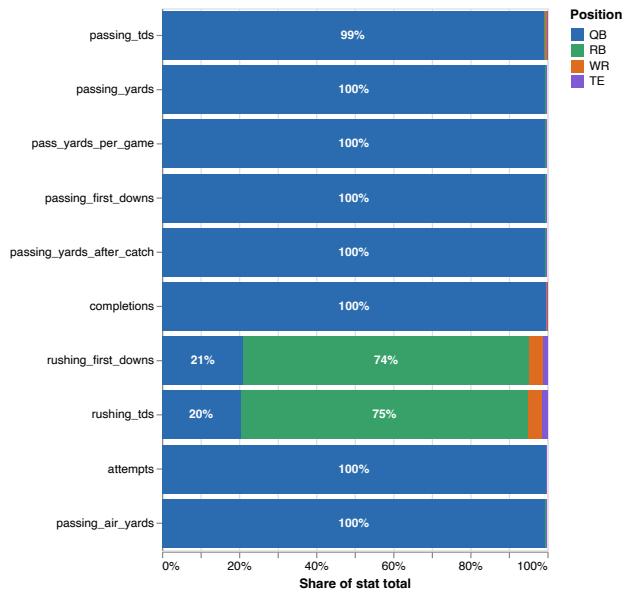
Scoring format clearly changes which positions matter most.

Football fans may already know which positions produce these stats, but several positions often contribute to the same category, so the chart below shows how each stat's total is split by position.

Top 10 Stats by Correlation: Which Position Produces Them
PPR



Non-PPR



Key Finding #1

The Most Important Statistics by Scoring Format

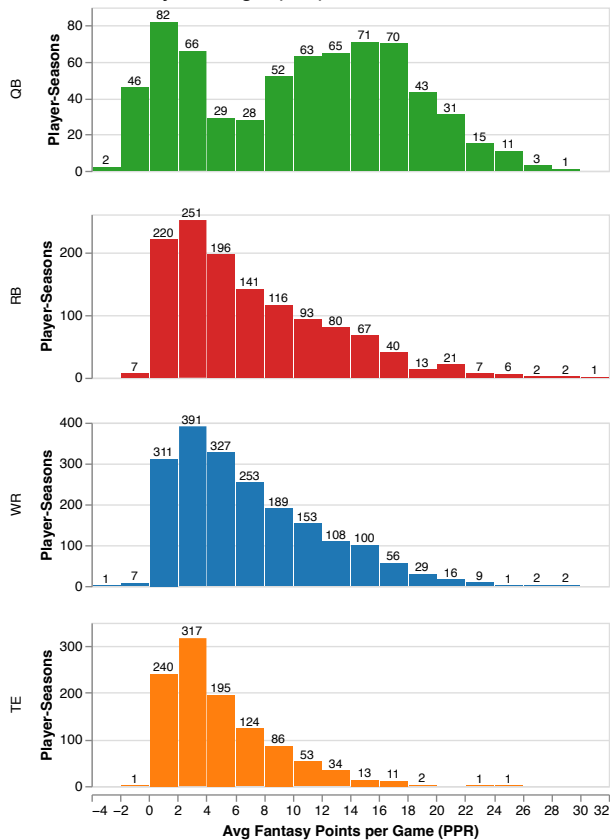
PPR: PPR players should focus on targeting the Wide Receiver position in drafts, as they own the largest share of 9/10 statistics with the highest correlations to fantasy points per game.

Non-PPR: Non-PPR players should focus on targeting the Quarterback position in drafts, as they own the largest share of 9/10 statistics with the highest correlations to fantasy points per game.

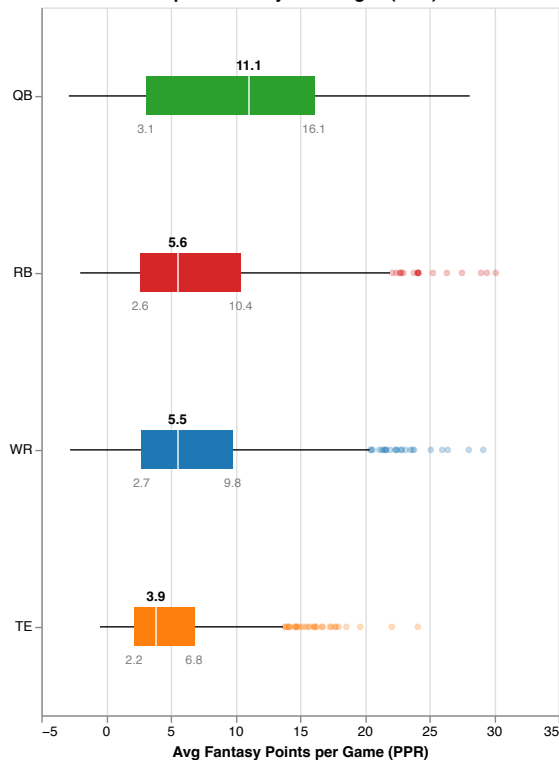
Now that we know which position is the most important, we need to rank all positions in order of performance. This will provide key strategic insights into how players of each scoring type should prioritize position groups.

Position-Specific Drivers

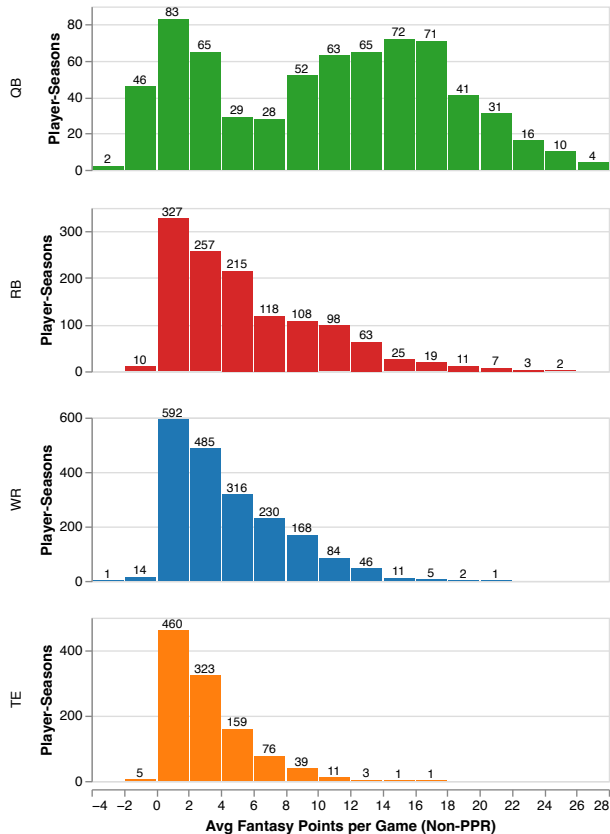
Average Fantasy Points per Game by Position (PPR)
Distribution of Player Averages (PPR)



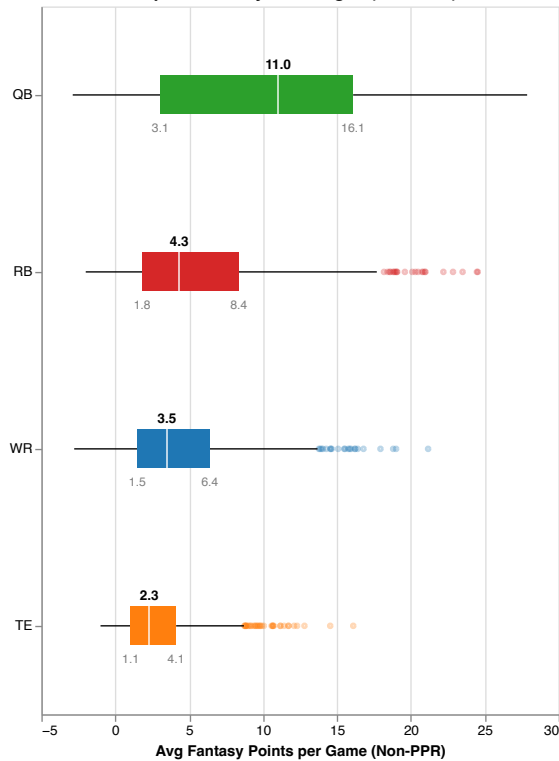
Spread of Player Averages (PPR)



Average Fantasy Points per Game by Position (Non-PPR)
Distribution of Player Averages (Non-PPR)



Spread of Player Averages (Non-PPR)



Early patterns in positional importance

Both scoring formats

1. **QBs score the most.** Their median points per game is consistently higher than any skill position (RB, WR, TE).
2. **Usable QBs are plentiful.** QB averages follow a roughly normal distribution, so many quarterbacks deliver solid, consistent production. Because a starter is easy to find, drafting a QB early is likely less important than filling other positions.
3. **Good TEs are scarce.** TEs have the lowest averages, and few exceed 6 points per game. That makes most TEs a low-impact pick, and the rare productive ones especially valuable.

PPR

- Skill positions score more than in non-PPR, which narrows the gap with QBs. With starting QBs easy to find, **skill-position picks matter most.**

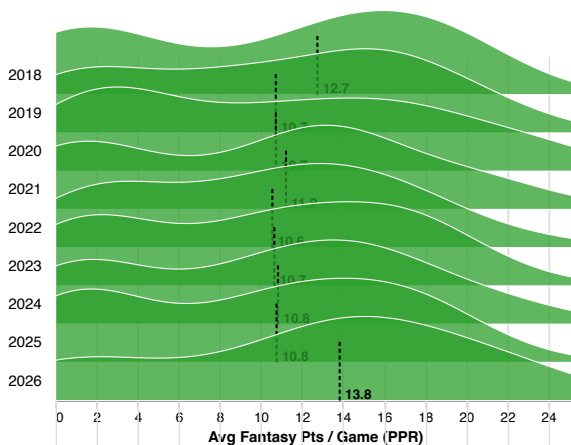
Non-PPR

- Skill positions average about 1.5 fewer points per game than in PPR, which widens the gap with QBs. **An elite QB may be more valuable in this format**, though further analysis is needed to confirm it.

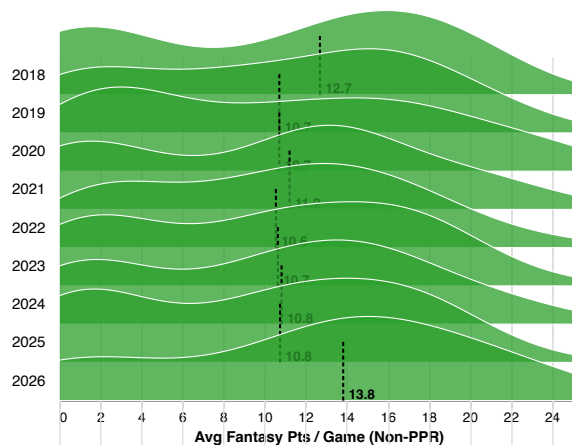
Before ranking positions by importance, we need to see if they are consistent over time.

Are these findings tried and true patterns, or outlier driven?

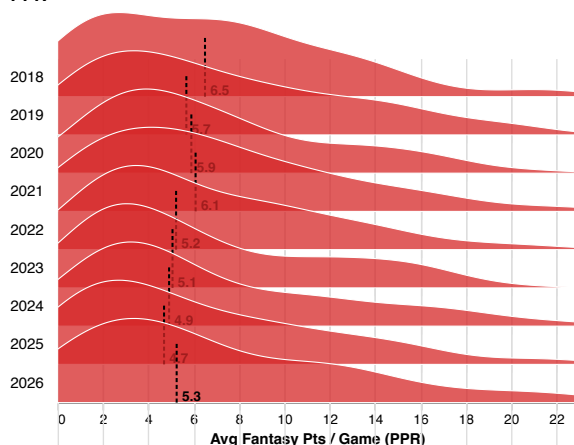
QB: Distribution of Avg Fantasy Points per Game by Season
PPR



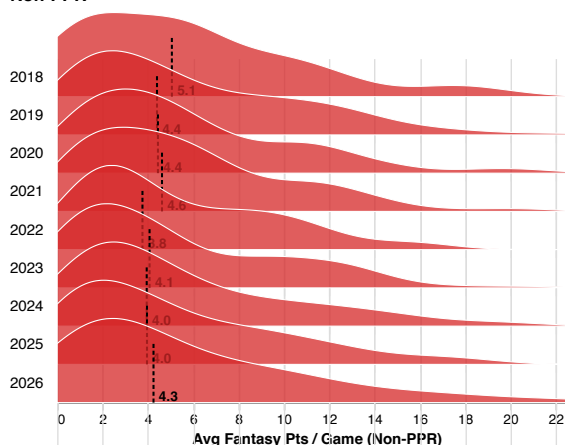
Non-PPR



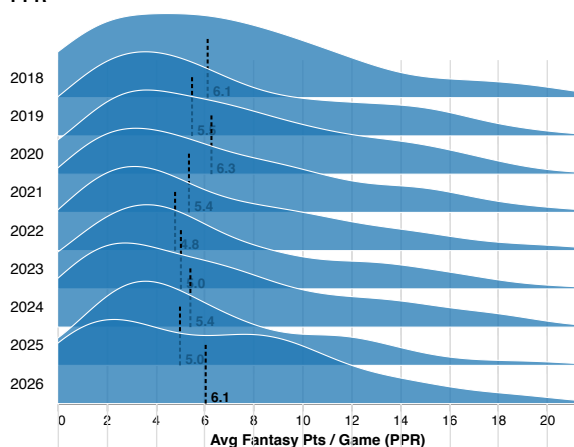
RB: Distribution of Avg Fantasy Points per Game by Season
PPR



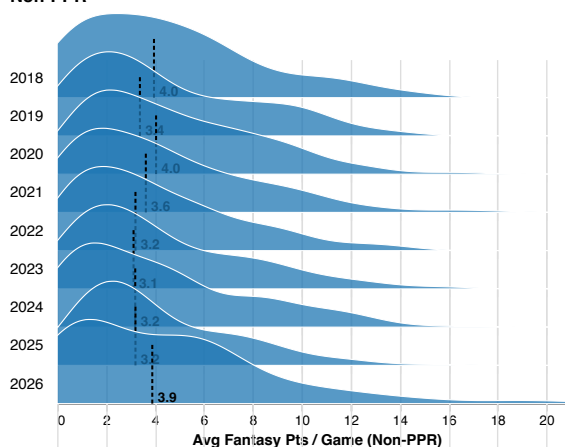
Non-PPR



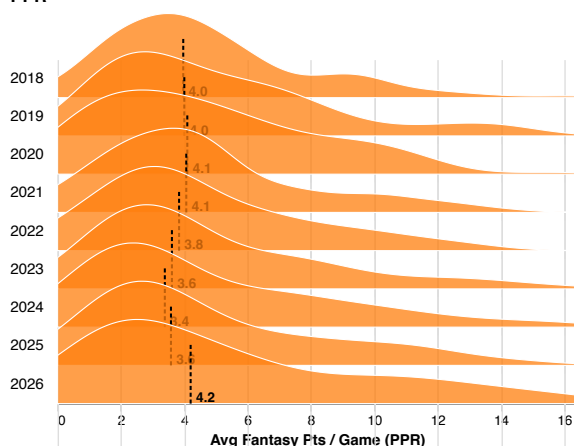
WR: Distribution of Avg Fantasy Points per Game by Season
PPR



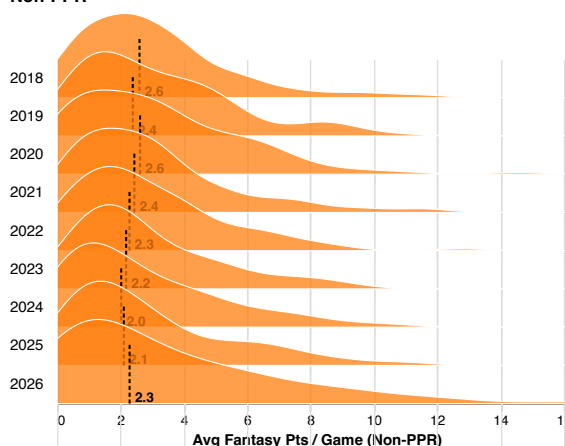
Non-PPR



TE: Distribution of Avg Fantasy Points per Game by Season
PPR



Non-PPR



Key Finding #2:

Positional Importance Differs by Scoring Format

Relevant to Both Formats

- **RBs and WRs outscore TEs.** TEs are tied to more of the high-correlation stats as their offensive responsibilities similar to Wide Receivers , but RBs score more points per game.

- **Productive RBs are easier to find than productive TEs.** More RBs make consistent contributions, so drafting a useful RB is more likely than drafting a useful TE.
- **QBs provide consistent scoring, but are plentiful.** They have the highest average points per game, but the distribution indicates that there are no shortage of safe options.

PPR

- **WRs produce most of the stats that drive scoring.** They account for the largest share of the stats most correlated with points per game.
- **WRs and RBs lead the skill positions in scoring.** Their average points per game are roughly tied.
- **WRs have a higher floor than RBs.** Their 25th-percentile scoring is higher, so even lower-end WRs contribute more reliably.

Non-PPR

- **QBs dominate the stats that drive scoring.** They hold the largest share of 9 of the 10 stats most correlated with points per game, making QB much more important in this format.

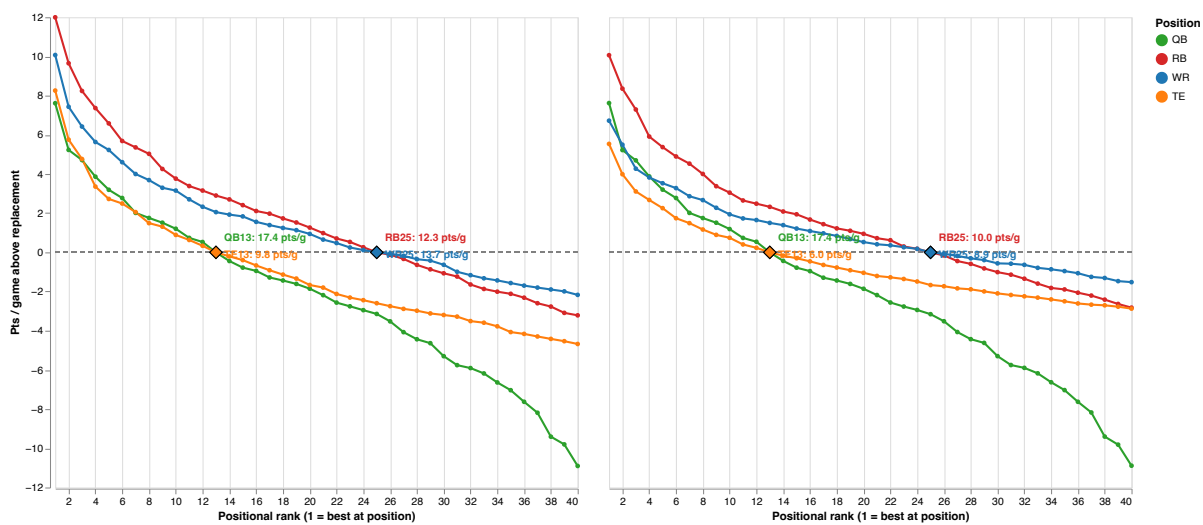
While this context is helpful, we want to go one level deeper and suggest a strategy around which positions to target first by scoring format, ranked by urgency.

Methodology

- **Data:** NFL regular-season weekly stats, 2018–2025 (2026 excluded as a partial season), limited to QB, RB, WR and TE players with at least 6 games in a season.
- **Per-game scoring:** each player's average fantasy points per game for the season, computed separately for PPR and non-PPR scoring.
- **Positional rank:** players are ranked within their position each season by points per game (1 = best), then the score at each rank is averaged across seasons. Each point therefore represents a *typical* Nth-best player, not a specific player.
- **Replacement level:** the first non-starter at each position, assuming a 12-team league, every team must start a specific number of players at each position. EX: If league has two wide receiver slots per team, and there are 12 teams, this would be the 25th ranked receiver.
- **Value over replacement:** each rank's points per game minus its position's replacement level. Higher values and steeper curves mean bigger gains from drafting that position early.

Positional Draft Strategy

Positional Drop-off: Points per Game Above Replacement (♦ = replacement level)



Key Finding #3: Ranking Positions by Draft Priority

Positions are ranked by how much the best player at each position outscores a replacement level player in points per game. A bigger gap means waiting on that position costs more.

PPR

1. **Running Back:** +11.7 pts/game
 - WRs produce the largest share of high-correlation stats, but RBs drop off more steeply, so elite RBs are harder to replace and should be drafted first.
2. **Wide Receiver:** +10.1 pts/game
3. **Tight End:** +8.3 pts/game
4. **Quarterback:** +7.6 pts/game

Non-PPR

1. **Running Back:** +10.1 pts/game
 - QBs produce the largest share of high-correlation stats, but RBs drop off more steeply, so elite RBs remain the top priority.
2. **Quarterback:** +7.6 pts/game
3. **Wide Receiver:** +6.7 pts/game
4. **Tight End:** +5.5 pts/game

This provides a strategic framework for fantasy players to approach how they should select their team.

While the blueprint is helpful, it doesn't help us to identify which players to take. Given the ethical concern we raised above, we will refrain from providing specific player recommendations and instead provide analysis to highlight general statistics that players use to choose optimal players at each position.

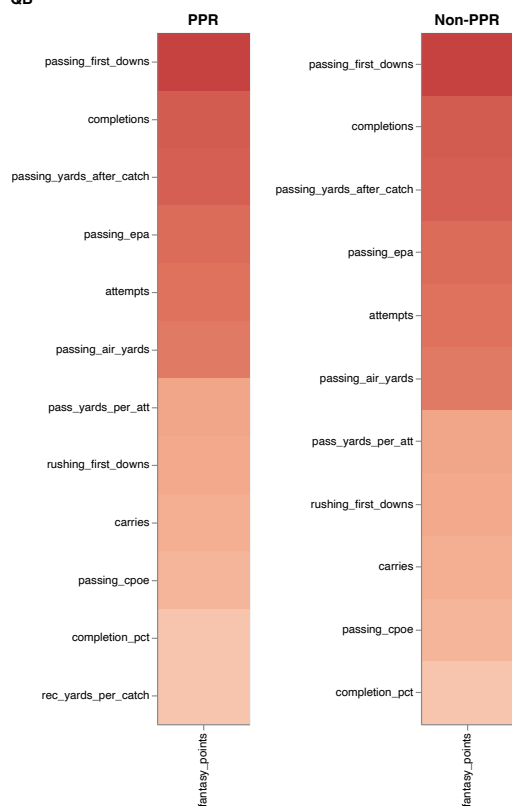
Player Selection by Posistion

Correlations

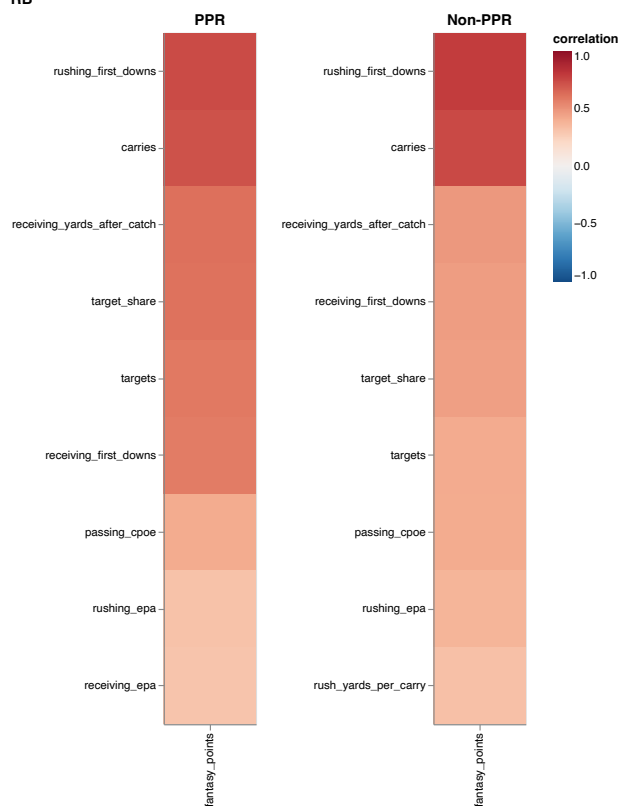
Only correlations with $|r| \geq 0.3$ are shown, to focus on the most impactful features.

Stats that are part of the scoring formula itself, such as receiving yards and touchdowns, are excluded. They correlate with fantasy points by definition, so including them would show what *produces* points rather than what *drives* a player to score them.

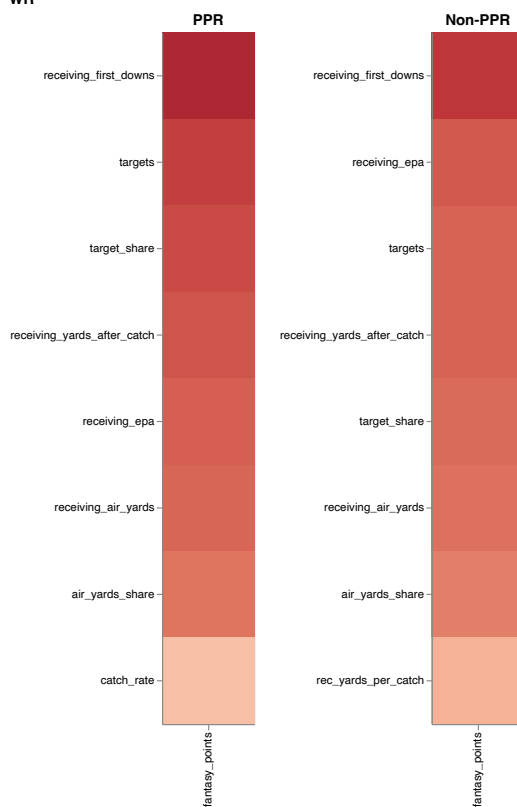
Stats Correlated with Fantasy Points by Position (lri ≥ 0.3)
QB



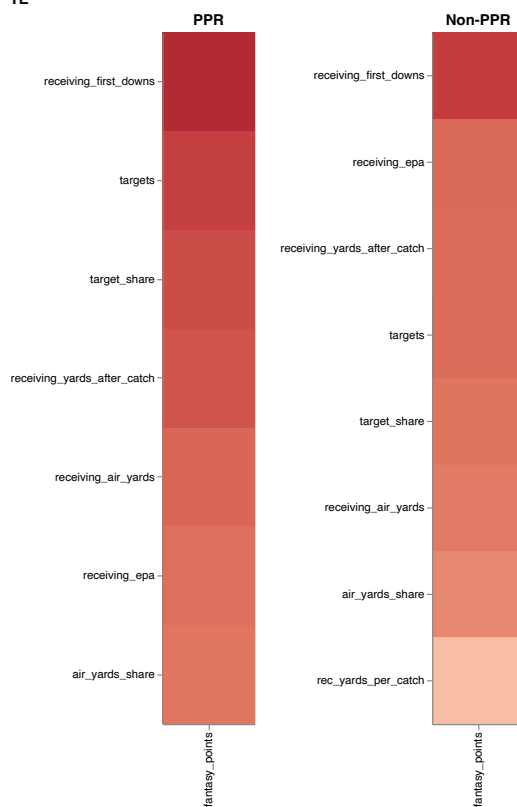
RB



WR

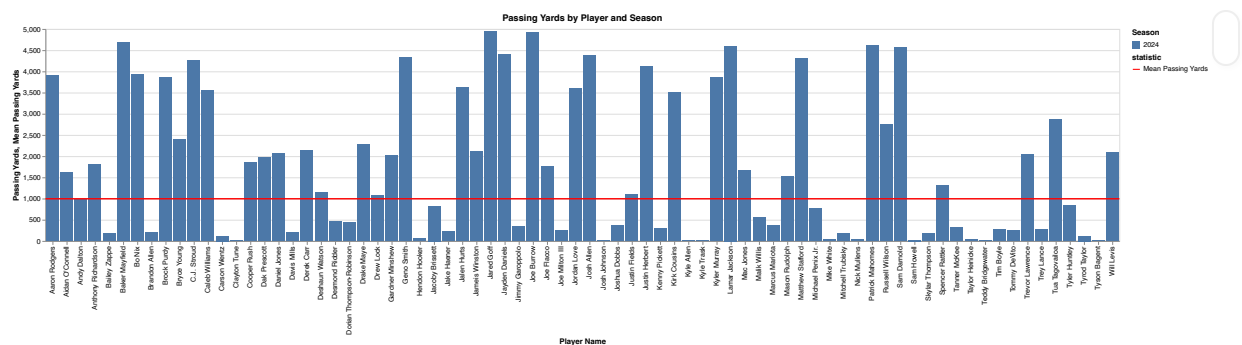


TE



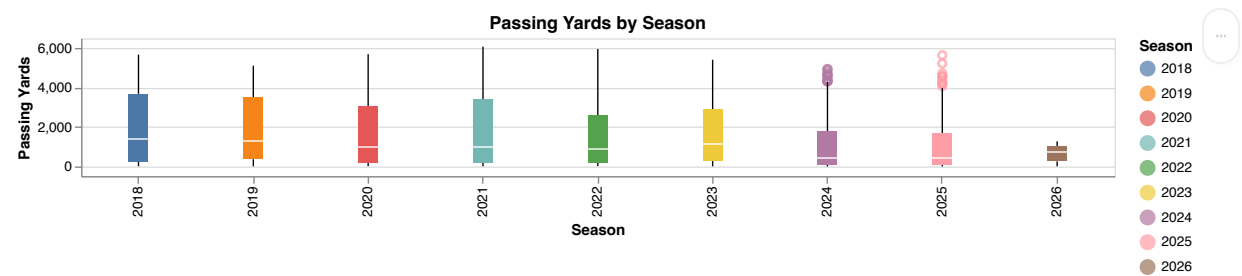
Quarterback

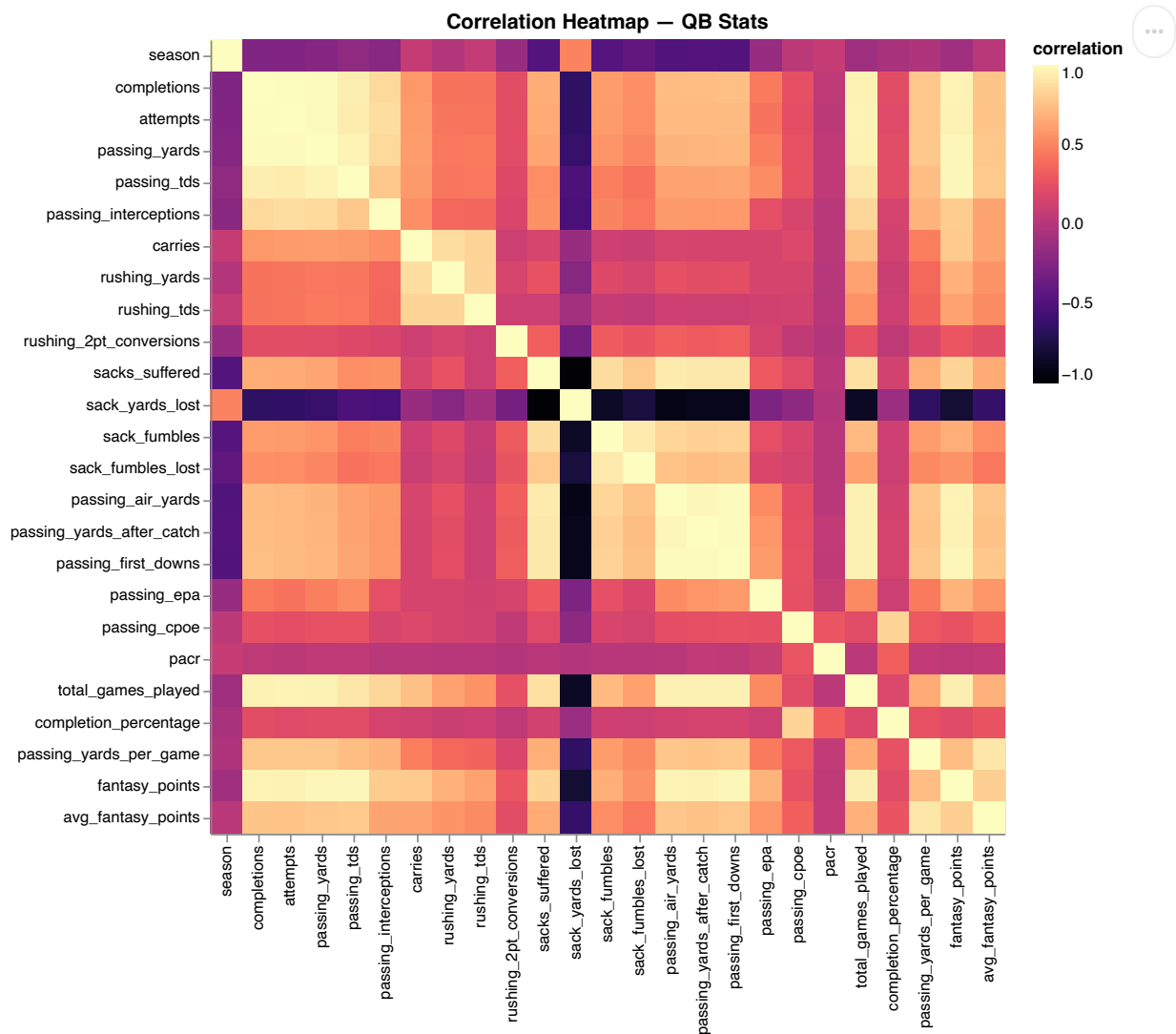
Create Filters



Season 2024

Data Source NFL

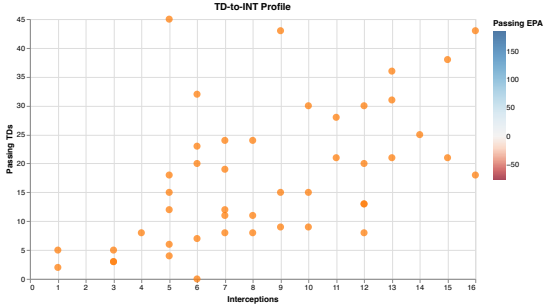
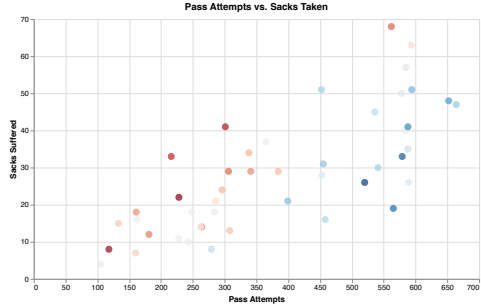
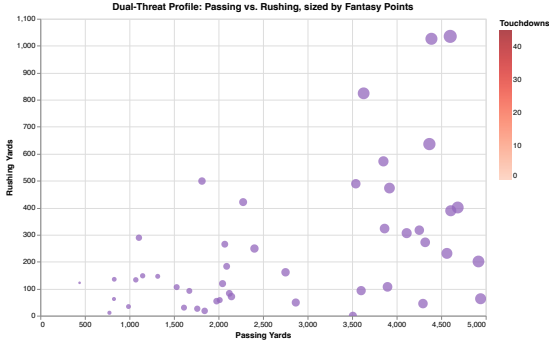
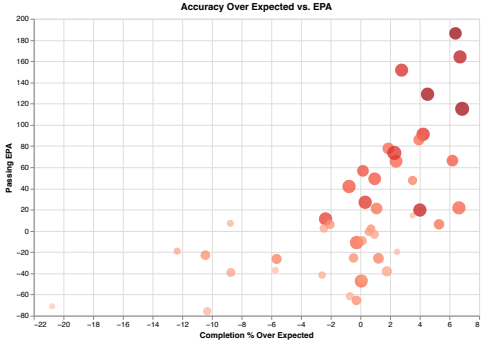
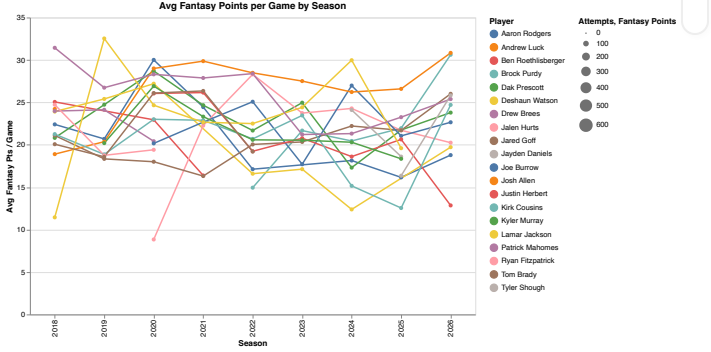
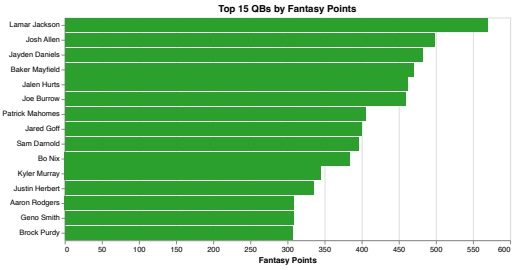




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Saved dashboard to qb_dashboard.html

QB Fantasy Football Dashboard

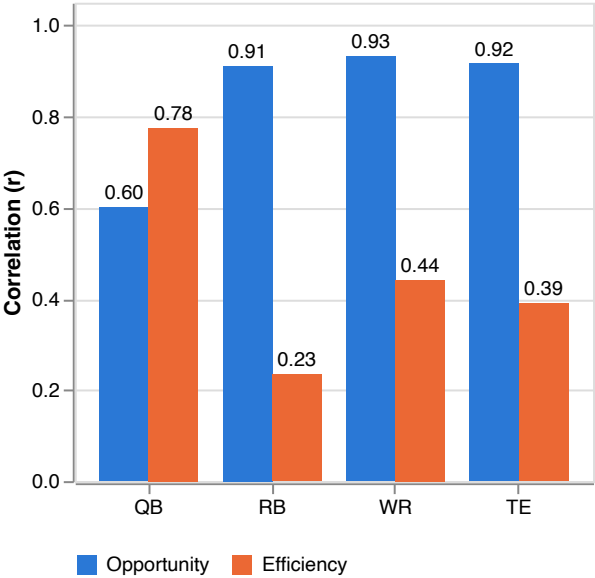


Season

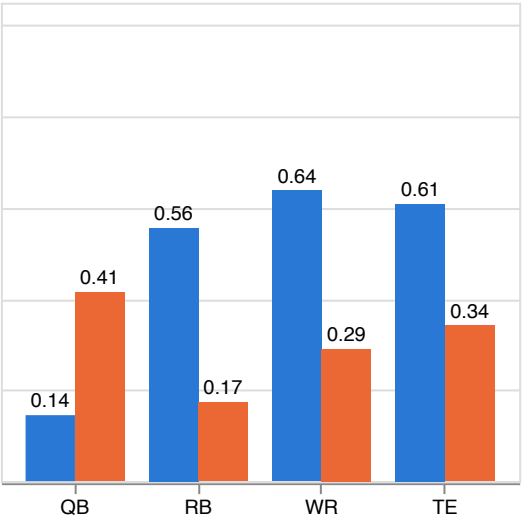
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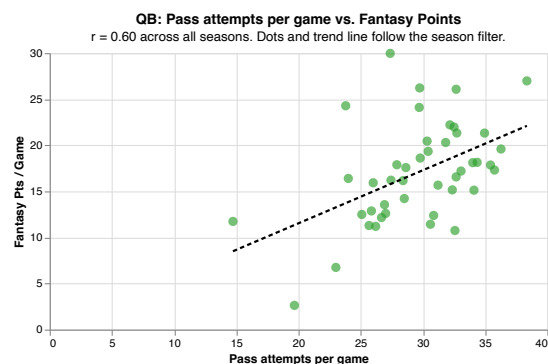
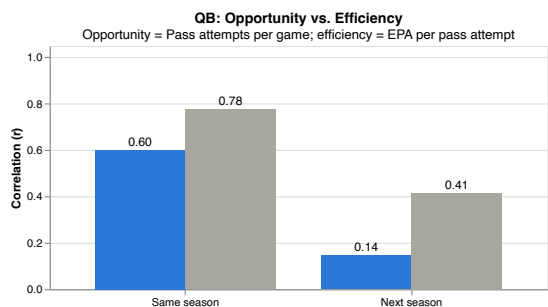
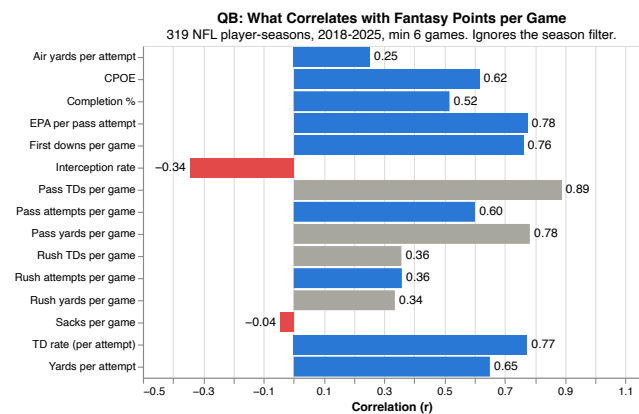
Opportunity vs. Efficiency by Position

Same season



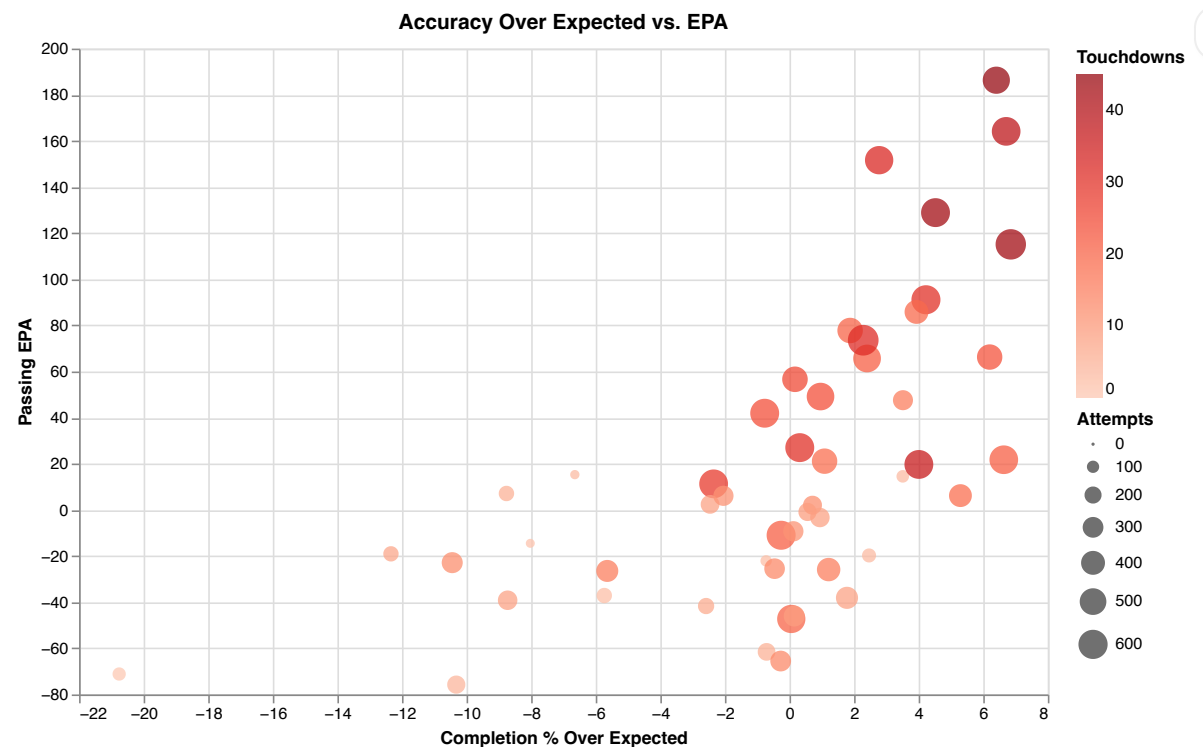
Next season





Season 2024

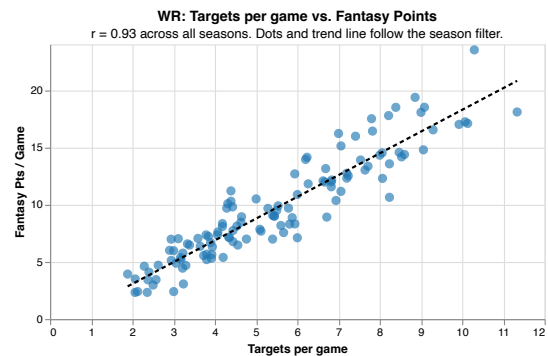
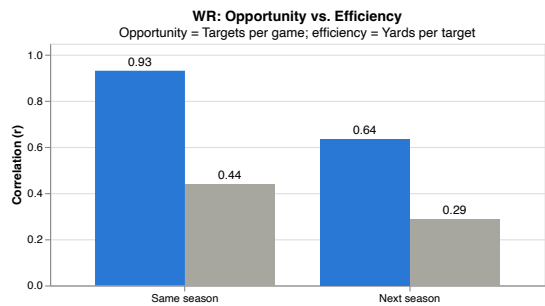
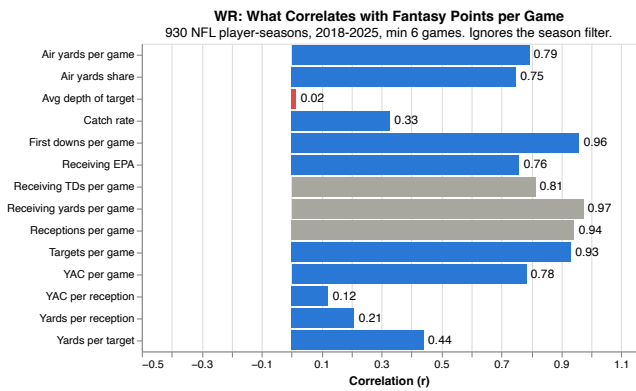
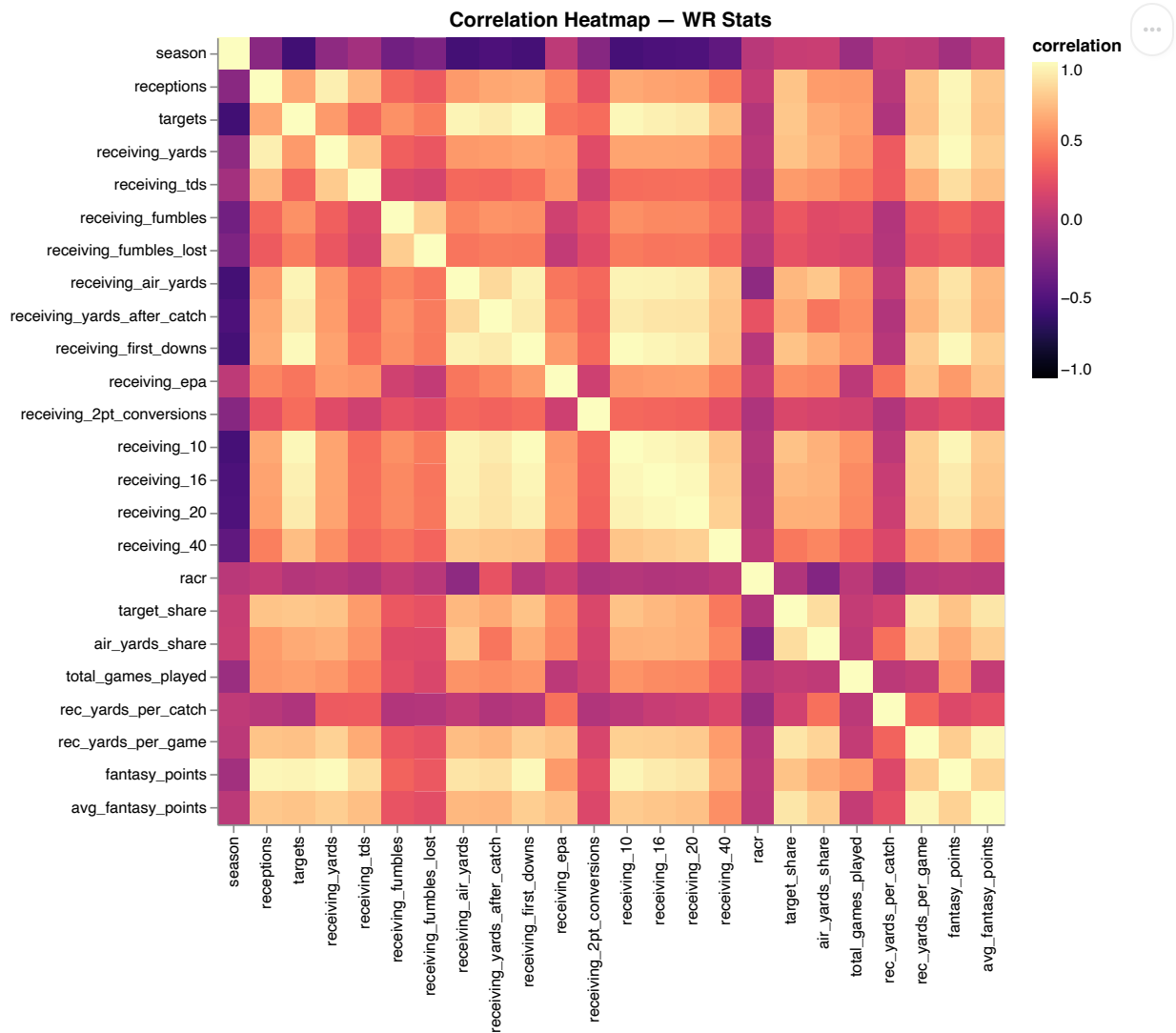
Underlying driver Scoring component Negative / none Opportunity Efficiency



Season 2024

Data Source NFL

Wide Receiver



Saved dashboard to wr_dashboard.html

WR Fantasy Football Dashboard



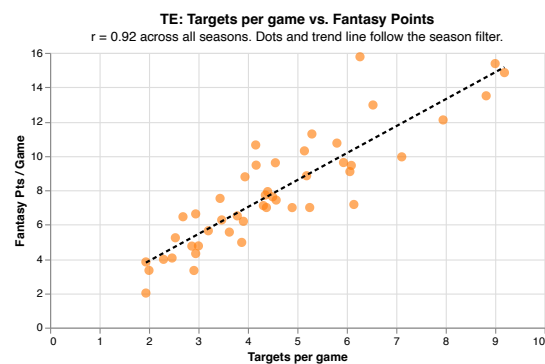
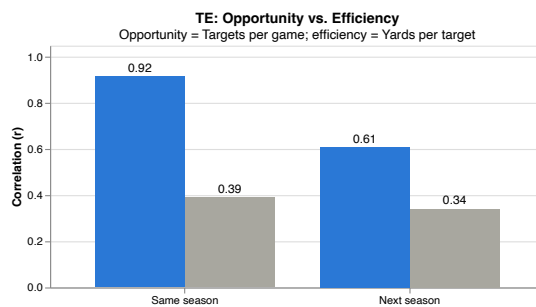
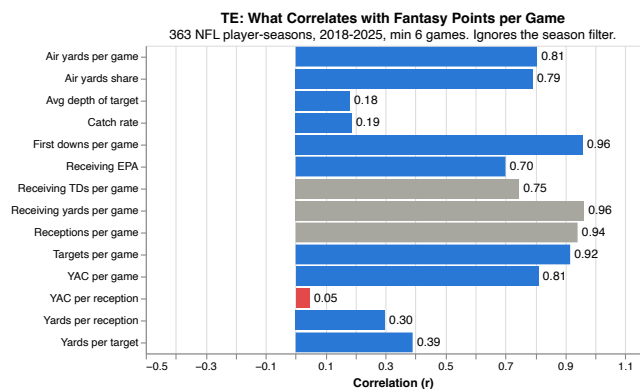
Season

Data Source

Air Yards vs Target Share

Target Share vs Receiving Yards

Tight End

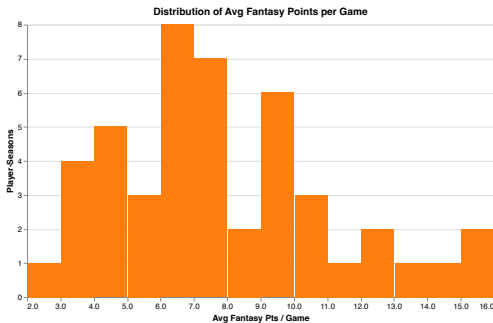
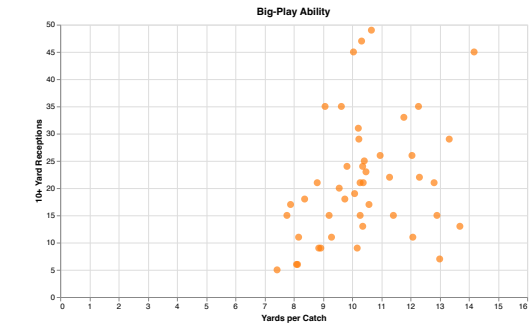
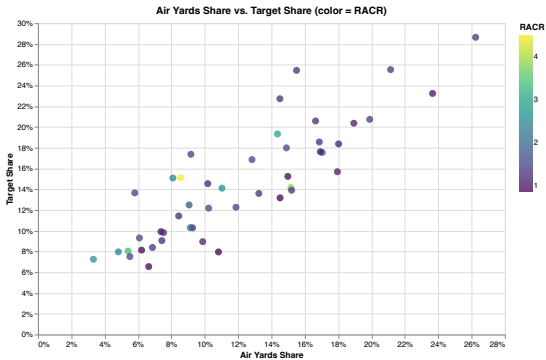
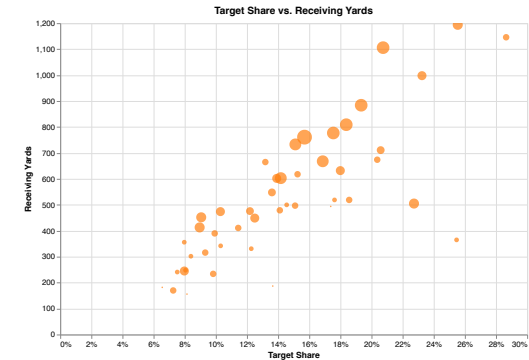
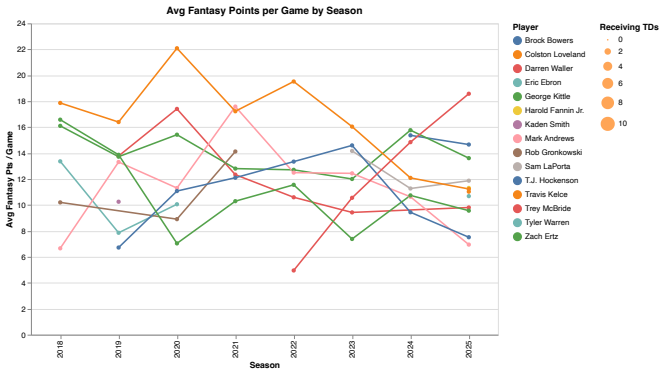
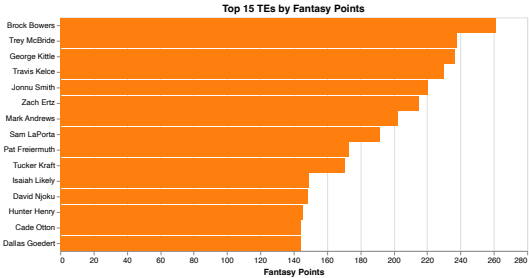


■ Underlying driver
 ■ Scoring component
 ■ Negative / none
 ■ Opportunity
 ■ Efficiency

Season 2024 ▼

Saved dashboard to `te_dashboard.html`

TE Fantasy Football Dashboard



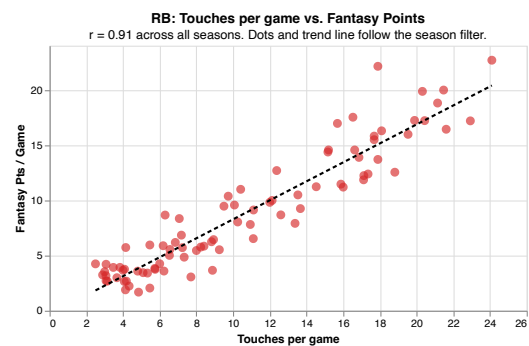
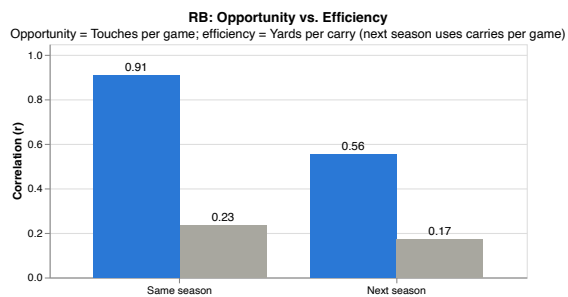
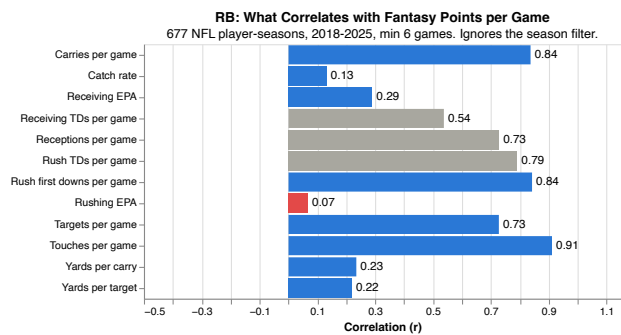
Season

Data Source

Air Yards vs Target Share

Target Share vs Receiving Yards

Running Back

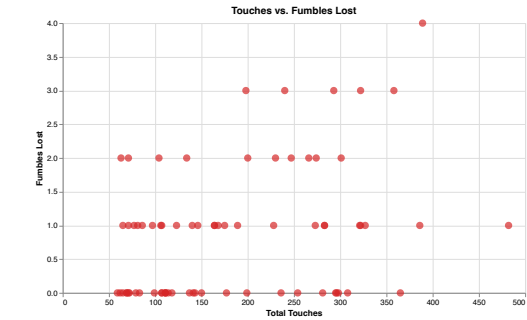
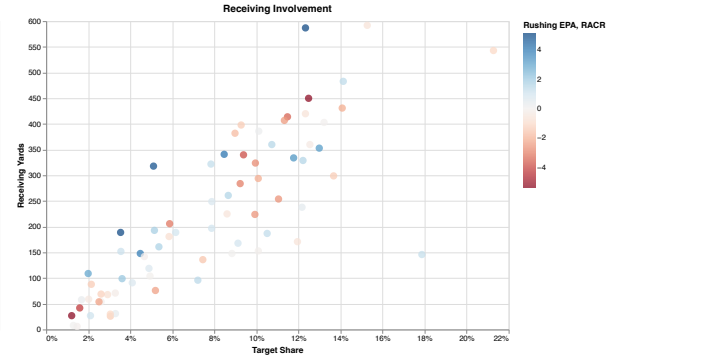
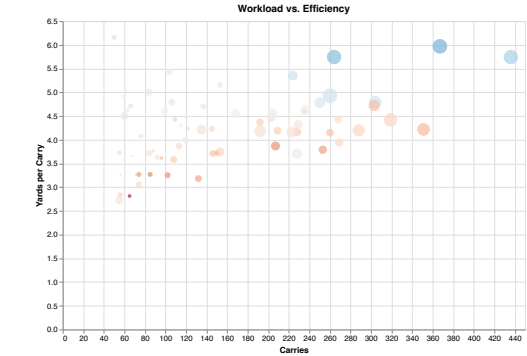
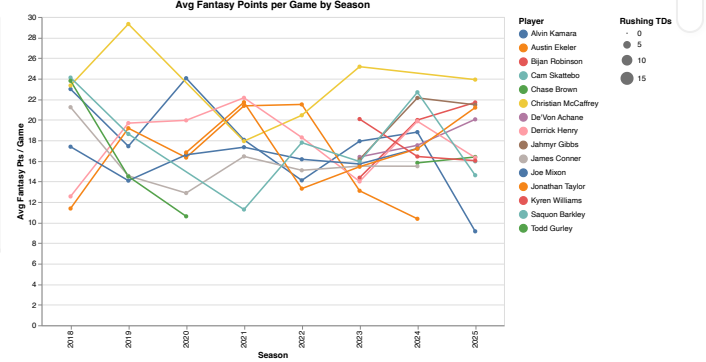
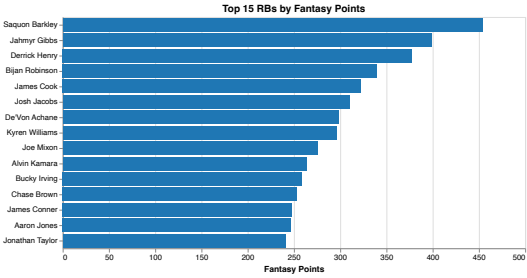


■ Underlying driver ■ Scoring component ■ Negative / none ■ Opportunity ■ Efficiency

Season 2024 ▼

Saved dashboard to rb_dashboard.html

RB Fantasy Football Dashboard



Season 2024

Data Source NFL